Studio North: an agentic AI customer-data lab

Are We Moving in the Right Direction?

How eight customer datasets become an executive report

Managing Customer Data · Mod II 2026

Synthetic records, explicit methods, and a workflow you can inspect

How to read the slide colors

Concept	Definitions and distinctions that organize the analysis.
Case application	Studio North, its records and the report-building workflow.
Evidence interpretation	What the saved data, calculations and forecasts establish.
Group decision	A judgment about the business or a prompt for further work.

The lab connects the course into one analysis

The agent carries out the work; you inspect the evidence and make the management judgment.



UCV and customer journey

Incrementality, cohorts, segmentation and LTV

Evidence, recommendation and reproducible appendix

One workflow links the customer information to a directional recommendation.

The business question gives the workflow its purpose

Studio North	A monthly creative-work membership for independent professionals.
Customer needs	Ready-to-use templates save time; guided learning builds skills.
Customer economics	Monthly renewals and add-on purchases produce contribution over time.

Are newer acquisition cohorts creating more value relative to their acquisition cost?

The customer-history exports describe different events

One row means something different in each source. Start by identifying that row unit.

Source table	Rows	What one row records
people.csv	5,703	One operational customer
subscription_periods.csv	43,082	One scheduled subscription cycle
transactions.csv	52,605	One payment or refund event
product_events.csv	69,590	One product or support interaction

Frozen synthetic exports · January 2024–December 2025 · 5,703 operational customers

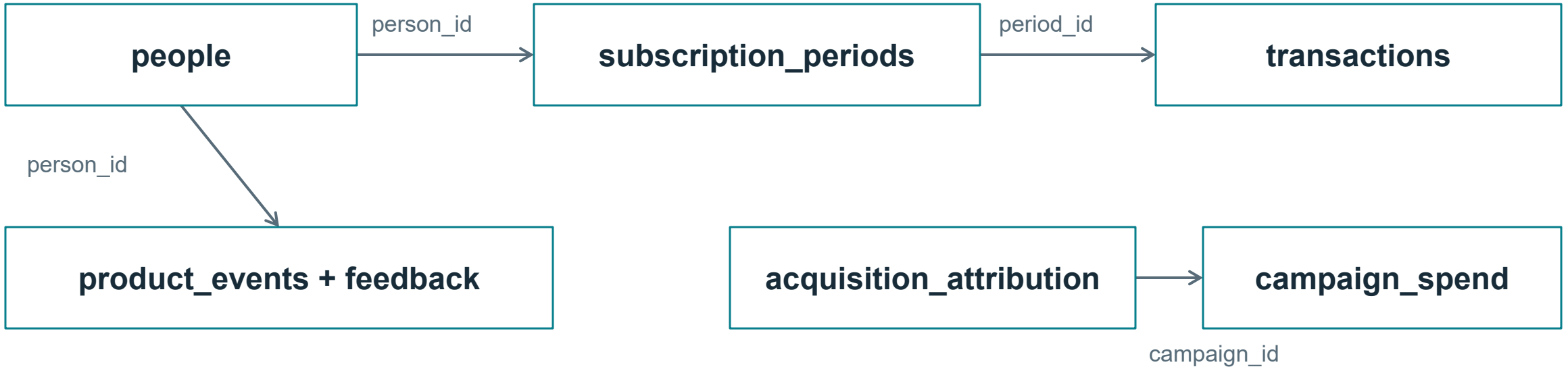
Four more exports connect experience and acquisition

The acquisition experiments are a separate research sample with their own controls.

Source table	Rows	What one row records
feedback.csv	3,321	One completed linked survey
campaign_spend.csv	288	One campaign on one spending date
acquisition_attribution.csv	5,703	One acquired person and source credit
channel_holdouts.csv	48	One randomized arm in one monthly test

All eight CSVs together: 20.89 MB uncompressed · 48 holdout rows = 24 tests × 2 arms

Keys connect records; row units determine valid joins



Attribution links person_id and the first payment.
Aggregate campaign spend before joining.

channel_holdouts stays separate: join its channel-and-need estimates only at the cost/value finale.

The Universal Customer View classifies fields

people.csv

Identifiers	Engagement	Relationship	Perception
person_id age_band household_type region	—	record_created_at attributes_recorded_at	primary_need

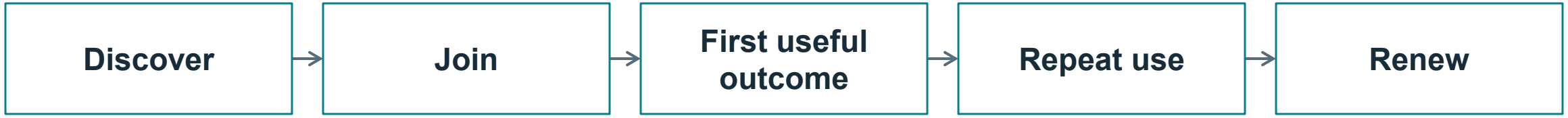
feedback.csv

Identifiers	Engagement	Relationship	Perception
response_id person_id	submitted_at journey_stage	—	ease_rating value_rating main_difficulty

The workflow generates a complete four-column field table for every source. Missing perceptions remain missing.

The journey map connects customer goals to evidence

One broad customer journey makes the service concrete before we analyze its records.



Evidence	What it contributes
Product events	Observed usage and support interactions
Feedback	Reported ease, value and difficulties
Renewal records	Whether the paid relationship continues

Potential friction is a hypothesis; pre-renewal behavior and feedback supply the diagnostic evidence.

The assets folder makes methodological sources inspectable

Source or asset	Role in this report
Fader and Hardie (2007; 2010)	Retention heterogeneity and customer-base valuation
Orduz (2024)	Gamma revenue structure; a separate neural retention benchmark
Larsen (2019)	Background acquisitions, incrementality and acquisition cost
Archify, report styles and locks	Diagrams, document design and a reproducible software environment

The licensed Orduz notebook is bundled with its source, license and hashes. Larsen has a source record and link.

Methods are adapted explicitly. The source assets explain what was borrowed and what the lab changed.

Three skill files have distinct responsibilities

Skill	Responsibility
SKILL.md	Defines the analysis, economic measures, evidence rules and outputs
skills/student-teacher/SKILL.md	Introduces the case, displays intermediate outputs and waits for “next step”
skills/executive-report/SKILL.md	Specifies the executive narrative, exhibits, citations and technical appendix

Skills state the instructions. SQL and Python compute the results. The report template turns those results into a document.

Choose a teaching walkthrough or a straight-through run

Guided class experience

Nineteen checkpoints. See actual intermediate tables and charts, discuss the result, then say “next step”.

Straight-through run

An explicitly requested batch run completes the analysis, model fitting and report without teaching checkpoints.

In the guided version, your final verdict and proposed action are recorded before the report is generated.

Intermediate outputs make the analysis visible

Checkpoint	What the student sees
Sources and UCV	Eight inventories, relationships and field-level classifications
SQL and observed cohorts	Saved queries, one traced cell and cohort economics
Segmentation and acquisition	Comparable grouping evidence and six monthly iCAC waves
Model and value	The model explanation, validation and LTV/payback forecasts
Recommendation and report	The learner’s judgment, supporting evidence and final PDF

lesson_state.json preserves your place; learner_choices.json preserves your judgments.

SQL turns event records into a customer clock

First paid acquisition

Find each person’s first positive successful payment and assign the acquisition month.

One customer-month

Aggregate transactions and subscription status before joining them to the monthly customer grid.

Cohort and age

Group by joining month and months since acquisition; retain the original acquired-customer denominator.

Python pivots the saved SQL aggregate into the displayed cohort triangle.

One report cell can be traced to a saved table

January 2024 cohort · Original acquired customers: 179

Calendar month	Age (months)	Retained customers	Retained share
2024-01	0	179	100.0%
2024-02	1	143	79.9%
2024-03	2	126	70.4%

At age 2: 126 retained ÷ 179 acquired = 70.4%

Observed inactivity can be zero. Future, unobserved ages stay blank.

Segmentation seeks similarity within cells and differences across them

Compare the same 4,278 eligible customers at the same six-receipt horizon, using two groups per candidate.

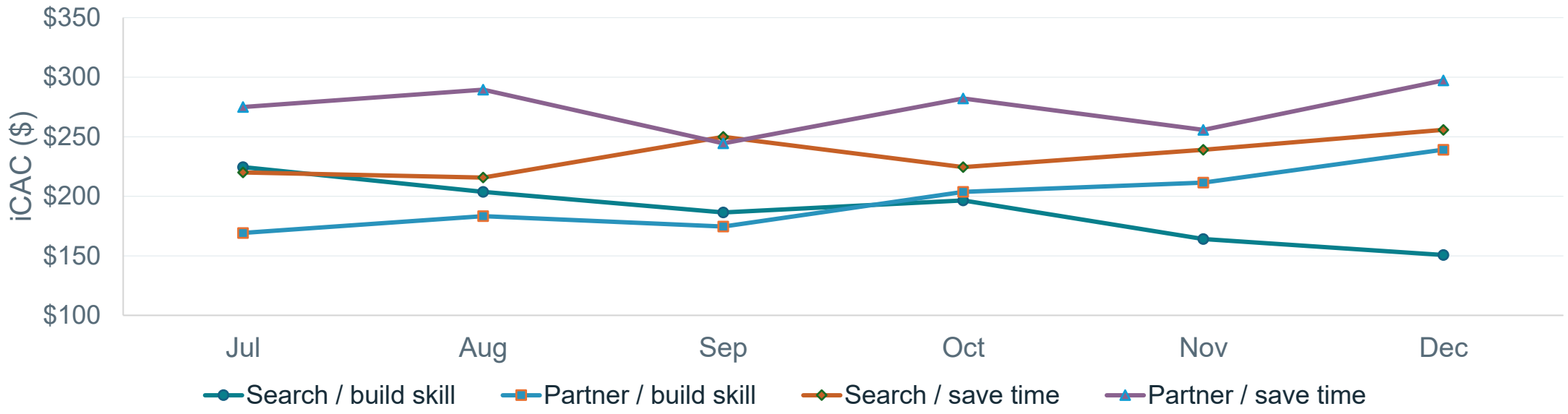
Grouping	Across-cell share of variation	Within-cell share
Stated need	8.3%	91.7%
Acquisition channel	0.8%	99.2%
Age band	0.0%	100.0%
Region	0.0%	100.0%

Stated need provides the strongest separation here; substantial variation remains within each group.

Outcome: contribution per acquired customer over the first six receipts.

Monthly holdouts keep acquisition costs current

Fresh holdout tests run each month in four channel-by-need cells. Each test follows 500 treatment and 500 control prospects for 31 days.



July–December 2025 point estimates. Latest completed tests update iCAC; intervals are in the report appendix.

A holdout reconciles credited CAC with iCAC

December 2025 · Search / build skill · Results complete January 1, 2026

Evidence	Treatment	Holdout
Assigned prospects	500	500
First paid buyers	112	39
Focal acquisition spending	$11,000	$0

Credited CAC = $11,000 ÷ 112 = $98.21

iCAC = $11,000 ÷ (112 − 39) = $150.68

The holdout measures background inflow from all other routes. Source: Larsen (2019).

The cohort model forecasts value from customer histories

Retention	Shifted Beta-Geometric: persistent differences in dropout risk, with cohort and group information.
Revenue and contribution	Gamma revenue among retained customers, combined with variable costs and observed contribution.
Chronological validation	Fit through September 2025, then check October–December. Compare with simple and neural benchmarks.

Experiments estimate acquisition cost. They contribute no retention or revenue outcomes to this model.

Sources: Fader and Hardie (2007; 2010); Orduz (2024)

The finale joins current acquisition cost with cohort value

Current iCAC comes from December tests. Value comes from the July 2025 operating cohort: six observed receipts, a common 12-receipt horizon.

Channel / need	Value before CAC	Current iCAC	Value less iCAC
Search / build skill	$284.03	$150.68	$133.35
Partner / build skill	$299.62	$239.13	$60.49
Search / save time	$186.66	$255.81	-$69.15
Partner / save time	$215.07	$297.30	-$82.22

Search / build skill is the leading candidate. Refresh iCAC as testing and spending continue.

Planning assumption: additional buyers have the matched cohort's contribution value. Acquisition cost is deducted once.

A recommendation has an inspectable evidence chain

Step	Saved evidence for Search / build skill
Source records	channel_holdouts.csv: 112 treatment buyers and 39 holdout buyers
SQL aggregation	channel_cac_current.csv: $150.68 per additional buyer
Customer-value forecast	Matched July cohort: $284.03 contribution value before CAC
Decision calculation	channel_value_decision.csv: $133.35 value less iCAC
Report exhibit	Named recommendation, calculation table and source map

An AI-written sentence can be checked against the files that support it.

Dependencies turn instructions into executable work

Component	What it does	How the version is preserved
Python + uv	DuckDB, data analysis, models and figures	pyproject.toml and uv.lock
Node + Chromium	Archify diagrams and browser exports	tools/package-lock.json
LaTeX + bundled styles	Compile the report into a PDF	Templates, fonts and build manifest
Source assets	Explain methods and supply licensed materials	Original source, license and hash records

A coding-agent harness handles the setup and runs the tools. Students guide the analysis through prompts.

Reproducibility preserves the work behind the PDF

Inputs and definitions	Frozen CSVs, documented keys, economic definitions and file hashes.
Executable analysis	Saved SQL, code, dependency locks, fitted outputs and source manifests.
Document and checks	Editable report source, figures, PDF, validation receipts and visual review.

A report replay reuses verified fits. A full run refits the models. Record which actually ran.

The appendix explains how the report was created

Where to look in the current report	What it explains
Main body: pages 1–15	The business diagnosis, supporting evidence and proposed action
Appendix begins: page 16	Technical detail that supports the management narrative
Dataset schema: page 37	The synthetic source tables, keys and relationships
Dependencies and assets: page 41	The software environment and supporting knowledge assets
Skill-driven workflow: page 42	How instructions, code, outputs and report generation connect

The final artifact is a report plus the evidence needed to review and reproduce it.

Reference List

Fader, Peter S., and Bruce G. S. Hardie (2007). “How to Project Customer Retention.” Journal of Interactive Marketing, 21(1), 76–90.

Fader, Peter S., and Bruce G. S. Hardie (2010). “Customer-Base Valuation in a Contractual Setting: The Perils of Ignoring Heterogeneity.” Marketing Science, 29(1), 85–93.

Orduz, Juan Camilo (2024). “Cohort Revenue Retention Analysis with Flax and NumPyro.” Licensed source-notebook edition included in assets/.

Larsen, Kim (2019). “You Don’t Know Your Customer Acquisition Cost.” Source record and original article link in assets/.

FAQ: What Else to Prompt after Running the Skill?

The report is a starting point for further questions.

Ask the agent to explain a transformation, change the view, compare a model or test a different decision.

The following seven prompts are editable and ready to copy.

FAQ 1: Describe the datasets through the UCV

Describe every dataset and provide a data dictionary for each. Explain its row unit, keys, links to other tables, and each field's meaning, type, units and missingness. Organize every field using the Universal Customer View: Identifiers, Engagement, Relationship and Perception. Explain ambiguous or contextual classifications and identify information we do not observe.

Copy the prompt into the same conversation after completing the report.

FAQ 2: Walk through the SQL cohort triangle

Walk me through the SQL-based process that turns individual-level transactional data into the report's triangular cohort view. Use a few actual customers, show the SQL and small intermediate tables, and explain each step: identify first paid acquisition, assign the acquisition cohort, calculate months since acquisition, build customer-month records, and aggregate cohort outcomes. Show how retention uses subscription records and the original acquired-customer denominator. Then explain how the aggregated results become the displayed triangle, why recent cohorts have fewer observed periods, and why future unobserved cells stay blank while observed months with no activity can be zero. Trace one displayed cell back to its source records.

Copy the prompt into the same conversation after completing the report.

FAQ 3: Compare four segments within each cohort

Reorganize the cohort-return charts so each figure represents one acquisition cohort and shows all four channel-by-need segments. Use consistent segment colors, axes and horizons, and distinguish observed returns from forecasts. Keep the underlying calculations unchanged. Explain which segments perform better within each cohort and how those differences evolve across cohorts.

Search / build skill · Search / save time
Partner / build skill · Partner / save time

FAQ 4: Research and test alternative LTV models

Research other LTV models, emphasizing unobserved heterogeneity within cohorts and/or customer-level LTV estimates. Explain what heterogeneity Longxiu's current shifted-Beta-Geometric model already captures and what each alternative adds. Propose train, validation and holdout splits, justify them, then compare the current approach with suitable alternatives using only information available at each forecast date. Evaluate LTV accuracy, calibration and implications for decisions on a common economic basis.

Copy the prompt into the same conversation after completing the report.

FAQ 5: Find a more useful segmentation strategy

Compare alternative segmentations using the available customer attributes. Seek within-cell homogeneity and across-cell heterogeneity in customer economics, comparing strategies at similar complexity. Evaluate held-out performance, segment size, stability and actionability, accounting for cohort differences. Does any strategy improve meaningfully on our current channel-and-need groups, and what different management action would it support?

Copy the prompt into the same conversation after completing the report.

FAQ 6: Identify what could change the recommendation

Stress-test the acquisition recommendations against alternative valuation horizons, discount rates, contribution assumptions and iCAC estimates. Allow additional customers' value to differ from the current cohort benchmark. Show which conclusions remain stable, what would reverse the preferred channel or cash-recovery assessment, and which additional evidence would most reduce that uncertainty.

Copy the prompt into the same conversation after completing the report.

FAQ 7: Adapt the skill to another business

Using a business description and sample exports I provide, show how to adapt this workflow to another company. Map its data to the required customer information, identify missing inputs, and explain which definitions, models, assets and skill instructions must change. Propose a reproducible implementation and validation plan that preserves the original Studio North case for comparison.

Copy the prompt into the same conversation after completing the report.